Import Tariffs and the Systematic Response of Monetary Policy*

Alessandro Franconi Lukas Hack

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Abstract

We estimate the macroeconomic effects of U.S. import tariff shocks using vector autoregressions and short-run restrictions. Tariff shocks reduce output but increase consumer prices. Monetary policy partially accommodates these shocks with a policy easing. To quantify the dependence on systematic monetary policy, we use empirically identified monetary policy shocks to construct counterfactuals that are robust against model misspecification and the Lucas critique. When monetary policy strictly stabilizes inflation, the output contraction at the trough is 36% larger than in the baseline. In contrast, strict output stabilization implies a peak inflation effect that roughly doubles, compared to the baseline.

Keywords: tariffs, trade, imports, monetary policy, counterfactuals
JEL Codes: C32, E31, E32, E52, F14

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1 Introduction

What are the macroeconomic effects of tariff shocks? How are these effects shaped by the systematic response of central banks? Partly motivated by the 2025 rise in tariffs, a myriad of papers has emerged to answer these questions, primarily using structural models. These papers study tariffs focusing on business cycle effects (e.g., Antonova, Huxel, Matvieiev, and Müller, 2025; Auclert, Rognlie, and Straub, 2025; Kalemli-Özcan, Soylu, and Yildirim, 2025; Costinot and Werning, 2025), tariff design (Becko, Grossman, and Helpman, 2025; Dávila, Rodríguez-Clare, Schaab, and Tan, 2025; Itskhoki and Mukhin, 2025; Kocherlakota, 2025), and the optimal monetary policy response (e.g., Bergin and Corsetti, 2023, 2025; Bianchi and Coulibaly, 2025; Monacelli, 2025; Werning, Lorenzoni, and Guerrieri, 2025).[1] While the above papers focus on theoretical analyses, there is little empirical evidence on the business cycle effects of tariffs, and no evidence on how the effects of tariffs are shaped by systematic monetary policy. We fill this gap.

In the first part of our analysis, we estimate the effects of import tariff shocks using a vector autoregression (VAR) and quarterly U.S. data from 1990 to 2024. Import tariffs are measured as the trade-weighted average tariff rate. The responses are identified via the timing restriction that import tariff shocks affect macroeconomic outcomes with a one-period lag, except for the federal funds rate.[2] We impose this assumption to allow other macroeconomic shocks to affect our import tariff measures through compositional changes, e.g., in import prices, imported products, and trading partners.[3]

Following this approach, we present robust evidence that persistent tariff shocks are contractionary and inflationary. This implies that tariffs act as supply shocks that pose a meaningful monetary policy tradeoff. Quantitatively, a tariff shock that increases the trade-weighted U.S. import tariff rate by 1 percentage point on impact leads to a significantly elevated

[1] Alessandria, Ding, Khan, and Mix (2025) focus on tariff revenues.

[2] We provide a comprehensive assessment of the robustness of these results in Section 5.

[3] For example, since the U.S. tariff code is regressive (Acosta and Cox, 2019), changes in income inequality and associated changes in import composition may affect trade-weighted tariff rates.

tariff rate for about 3 years. It raises CPI inflation with a peak effect of 0.78 percentage points after 11 quarters. The trough in real GDP is -1.23% and is reached after 6 quarters. This confirms the theoretical predictions that U.S. import tariffs are contractionary (Auclert et al., 2025) and act as supply shocks (Werning et al., 2025). Furthermore, we estimate an increase in macroeconomic uncertainty and a decrease in real investment, imports, and exports. The terms of trade improve, albeit with a delay.

Monetary policy partially accommodates the tariff shock with lower interest rates, potentially contributing to inflationary pressure while cushioning the decline in output. Interestingly, such partial accommodation can be the optimal response to import tariff shocks (e.g., Bergin and Corsetti, 2025; Bianchi and Coulibaly, 2025; Monacelli, 2025; Werning et al., 2025). The finding of partial accommodation raises several questions. How much of the inflation is genuinely caused by the tariff hike, and how much is due to the monetary easing? Similarly, how much output must be sacrificed to fully stabilize prices? And, conversely, how much inflation must be tolerated if the central bank fully stabilizes real activity?

To study these questions, we follow McKay and Wolf (2023) and use identified monetary policy shocks to construct policy counterfactuals. These counterfactuals are based on the monetary policy shock of Miranda-Agrippino and Ricco (2021) and the Taylor rule residual of Romer and Romer (2004).

First, we consider a counterfactual in which the federal funds rate is unresponsive to the tariff shock. This scenario is useful for comparison with the findings from Auclert et al. (2025), who also consider such a case. It also resembles the commonly expressed idea of "looking through" supply shocks. Interest rates are higher than in the baseline, leading to moderately lower real GDP and CPI inflation. Hence, we confirm that the inflationary impact of tariffs is partly driven by the monetary easing.

The above counterfactual keeps monetary policy neutral in terms of the nominal rate. But how potent is monetary policy in fighting the inflationary pressure from the tariff shock? To answer this question, we consider a counterfactual in which monetary policy aims to

perfectly stabilize prices. Instead of an initial easing, monetary policy sharply increases interest rates by 0.83 percentage points in the short run. Thus, the tariff shock does not lead to meaningful inflation, with a peak impact of only 0.22 percentage points.[4] The baseline inflation response in the same quarter is almost four times larger. However, this sacrifices a considerable amount of real GDP, which is 36% (0.44 percentage points) lower at the trough, when compared with the baseline.

Finally, we consider a counterfactual in which monetary policy aims to strictly stabilize output. In this case, monetary policy cuts the federal funds rate more aggressively and more swiftly than in the baseline. The trough interest rate response is -2.0 percentage points. This policy reduces the adverse output effects but does not achieve full output stabilization. Yet, the easing generates a pronounced amplification of inflation, with the peak effect being roughly twice as large as in the baseline.

Lastly, we provide several additional exercises. First, we discuss how key differences in the literature can partly be explained by the sample period under investigation. Second, we provide further results, including estimates of the fiscal footprint of tariffs. Finally, we present a comprehensive sensitivity analysis, including alternative identification and tariff measurement approaches, that confirms that our main results are remarkably robust.

Related literature. Beyond the theoretical papers mentioned above, our work is connected to three strands of literature. First, we relate to the surprisingly scant literature that identifies the macroeconomic effects of tariffs from aggregate data. Schmitt-Grohé and Uribe (2025) and den Besten, Barnichon, Känzig, and Singh (2026) find tariffs to be deflationary, which is partly driven by the sample period under consideration, as we discuss in Section 5.1. Different from the aforementioned papers, several earlier contributions find effects consistent with tariffs being supply shocks using sign restrictions with quarterly U.S. data (Boer and Rieth, 2024) and zero restrictions with annual cross-country data (e.g., Barattieri, Caccia-

[4]The counterfactual achieves inflation stabilization only approximately, since an exact counterfactual would require infinitely many distinct monetary policy (news) shocks.

tore, and Ghironi, 2021).[5] Relative to these papers, we contribute robust evidence from a more recent sample that better captures how tariff shocks propagate given the current economic structure. Moreover, we provide novel policy counterfactuals that allow us to better understand the monetary policy tradeoff posed by tariff shocks.

Second, a recent literature studies the tariff announcements by the Trump Administration on the so-called Liberation Day using structural models (e.g., Ignatenko, Lashkaripour, Macedoni, and Simonovska, 2025; Rodríguez-Clare, Ulate, and Vasquez, 2025) or high-frequency data (e.g., Acharya and Laarits, 2025; Jiang, Krishnamurthy, Lustig, Richmond, and Xu, 2025; Pinter, Uslu, and Smets, 2025; Yan and Morck, 2025). Such event studies provide complementary insights into the economic consequences of the trade war launched by the Trump Administration in 2025. Further related are Coibion, Gorodnichenko, and Weber (2025), who surveyed U.S. households about the potential Trump tariffs before the 2025 inauguration, and Cavallo, Llamas, and Vazquez (2025), who track the price impact of tariffs in real time.

Third, we relate to a literature on systematic policy changes. Using policy shocks, Barnichon and Mesters (2023) focus on the optimality of policy, whereas McKay and Wolf (2023) and Caravello, McKay, and Wolf (2024) concentrate on the construction of policy counterfactuals. In practice, such methods are applied to construct counterfactuals for monetary policy (Wolf, 2023; Franconi, 2024) and fiscal policy (Bouscasse and Hong, 2023; Breitenlechner, Geiger, and Klein, 2024). Different from these approaches, Hack, Istrefi, and Meier (2023) leverage the exogenous rotation of voting rights among policymakers.

[5]Further related work studies the financial market response to tariffs (Ostry, Lloyd, and Corsetti, 2025), trade policy uncertainty (e.g., Caldara, Iacoviello, Molligo, Prestipino, and Raffo, 2020; Poilly and Tripier, 2025), policy uncertainty (e.g., Baker, Bloom, and Davis, 2016), and geopolitical risk (e.g., Caldara and Iacoviello, 2022).

2 Data

This section introduces the data used in the baseline analysis and the additional data used for extensions and to investigate the sensitivity of the results. More details on the data are provided in Online Appendix A.

2.1 Main data for the baseline analysis

Average tariff rate. Our baseline tariff measure is the trade-weighted average U.S. import tariff rate, given by customs duties divided by dutiable imports.[6] The baseline sample period spans 1990Q1–2024Q4.[7] In Figure 1, we present the time series of this tariff rate as a solid line, along with the standardized tariff shock indicated by the bars, which we identify using the methodology explained in Section 3.

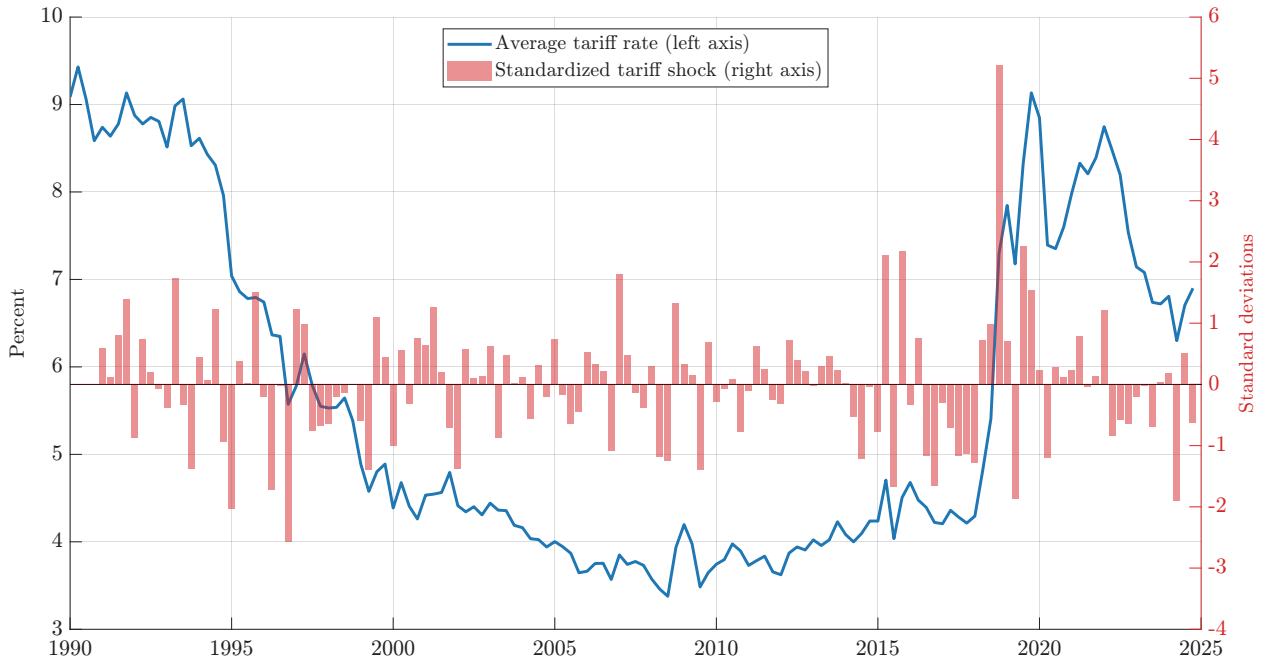
The average tariff rate fell during the 1990s and early 2000s, reflecting a general trend away from protectionism that also led to the NAFTA agreement in 1994 and the establishment of the WTO in 1995. This trend was sharply reversed with tariff increases due to the US–China trade war in 2018. These patterns suggest that there is meaningful variation in U.S. import tariffs in our sample. Our identified tariff shock picks up these important episodes: For example, we observe a large negative shock in 1995Q1 corresponding to the establishment of the WTO (the GATT Uruguay Round), and several positive tariff shocks during the 2018 trade war.[8] More generally, our identifying variation is well-distributed across the sample with sizable tariff reductions and increases.

[6] We use customs duties (FRED *B235RC1Q027SBEA*) divided by dutiable imports, given by all goods imports (FRED *A255RC1Q027SBEA*) multiplied by the average share of dutiable goods over all imported goods, computed using the DataWeb of the United States International Trade Commission (USITC).

[7] In Section 5, we discuss the role of the estimation sample in comparison to the related literature.

[8] Estimated responses to our tariff shock may also capture retaliation by third-party countries such as Chinese tariffs in direct response to U.S. tariffs.

Figure 1: Average tariff rate and the identified tariff shock



Notes: This figure shows the time series of the trade-weighted average U.S. import tariff rate as a solid blue line (left axis), and the standardized tariff shocks as red bars (right axis), which we identify based on the methodology explained in Section 3. The shock computed from the baseline Bayesian VAR by taking the mean of the reduced-form parameters.

Macroeconomic data. In addition to the tariff rate, our baseline analysis uses eight quarterly series: real GDP, year-over-year CPI inflation, and the federal funds rate (FFR) as measures of real activity, prices, and monetary policy, respectively. Beyond these core variables, we use real imports, real exports, and a terms-of-trade index to capture international trade dynamics. Lastly, we use real investment and macroeconomic uncertainty from Jurado, Ludvigson, and Ng (2015) to understand the effects of tariffs on the domestic industrial base.[9]

Monetary policy shocks. To analyze how monetary policy shapes the consequences of tariff shocks, we require multiple identified policy shocks. For our baseline analysis, we use the high-frequency-identified monetary policy shock from Miranda-Agrippino and Ricco (2021), which explicitly account for the possibility of Fed information effects, and the Taylor rule-type residuals from Romer and Romer (2004), extended by Wieland and Yang (2020).

2.2 Additional data for further results and sensitivity

Tariff measures. We consider three additional tariff series. First, we construct an unweighted average import tariff rate from disaggregated USITC data.[10] Second, we use the trade restrictiveness index of Feenstra (1995), computed at a quarterly frequency by Schmitt-Grohé and Uribe (2025). Third, we compile a series of 14 narratively identified tariff policy events. This series takes the value of the changes in the trade-weighted tariff rate for quarters with identified tariff events, and is zero otherwise.[11]

Macroeconomic data. To better understand the transmission of tariff shocks, we study real consumption, real imports of consumption goods, and the trade balance. To evaluate

[9] For data at higher frequencies, we take the arithmetic average within each quarter.

[10] First, we compute all disaggregated trade-weighted tariff rates at the four-digit HTS-by-origin-country level. Then, we compute the (unweighted) arithmetic average across these disaggregated tariff rates. To limit the influence of small trading partners, we drop all countries with average dutiable import values below the median.

[11] These events cover, e.g., the 2018 trade war and the establishment of the WTO. The full list of events and further details are in Online Appendix D.

the fiscal implications, we use customs duties, government spending, government receipts, and federal debt, all in real terms. We also use several alternative measures for the core variables. Instead of real GDP, we consider the unemployment rate, the CBO output gap, and industrial production. Instead of the CPI, we use the core CPI, the PPI, and one-year inflation expectations from the Michigan survey. As an alternative interest rate, we analyze the shadow FFR (Wu and Xia, 2016) as well as the one-year Treasury yield. Lastly, we use policy uncertainty indices (Baker et al., 2016), geopolitical risk (Caldara and Iacoviello, 2022), and trade policy uncertainty (Caldara et al., 2020), as well as the FRED-QD dataset (McCracken and Ng, 2021) and net-export SPF forecasts for diagnostic tests.

Monetary policy shocks. Finally, we use the heteroskedasticity-based shock from Jarociński (2024) and the refined Romer and Romer (2004) shock from Hack, Istrefi, and Meier (2024) to assess the sensitivity of our counterfactuals to the choice of monetary policy shocks.

3 Methodology

This section lays out the VAR model along with the identification of the tariff shock and the construction of monetary policy counterfactuals following McKay and Wolf (2023), MW henceforth.

3.1 Econometric methodology

VAR model. Our analysis is based on a covariance-stationary VAR model of the form

$$y_t = A(t) + B_1 y_{t-1} + \cdots + B_p y_{t-p} + u_t, \tag{1}$$

where $y_t \in \mathbb{R}^{n\times 1}$ is a vector of endogenous variables, $A(t) \in \mathbb{R}^{n\times 1}$ is a vector of deterministic components that may only depend on calendar time t, and $u_t \in \mathbb{R}^{n\times 1}$ is a vector of reduced-form innovations with zero mean and covariance matrix Σ, and $B_1, \ldots, B_p$ are coefficient

matrices with conforming dimensions. We estimate the VAR using conventional Bayesian techniques with inverse-Wishart priors on the reduced-form parameters.[12]

Causal effects. The reduced-form innovations are related to structural shocks $\varepsilon_t \sim (0, \mathbb{I})$ through $u_t = S\varepsilon_t$, where $S \in \mathbb{R}^{n\times n}$ is the impact matrix and the covariance matrix $\mathbb{I} \in \mathbb{R}^{n\times n}$ is the identity matrix, which is a standard normalization (e.g., Kilian and Lütkepohl, 2017). We assume the VAR to be invertible so that it admits a structural vector moving average representation

$$y_t = \sum_{h=0}^{\infty} \Theta^h \varepsilon_{t-h}, \tag{2}$$

with coefficient matrices $\Theta^h = (\theta_1^h, \dots, \theta_n^h)$ and $\theta_j^h \in \mathbb{R}^{n\times 1}$ denoting the causal effect of ε_t^j on y_{t+h}. We denote by $\theta_{i,j} = (\theta_{i,j}^0, \theta_{i,j}^1, \dots)$ the full response path of variable i to shock ε_t^j.

3.2 Counterfactual methodology

Target counterfactual. The causal effects of macroeconomic shocks, including tariff shocks, are generally a function of (unobserved) parameters $\mathcal{P} \in \mathbb{R}^{k\times 1}$ that describe the conduct of monetary policy, i.e., $\theta_{i,j} \equiv \theta_{i,j}(\mathcal{P})$. These parameters can encode a policy reaction function, but also a targeting rule that the monetary authority optimally follows. We are interested in the causal effects of tariff shocks under counterfactual conduct of monetary policy, denoted by $\widehat{\mathcal{P}} \neq \mathcal{P}$ and the corresponding causal effect $\widehat{\theta}_{i,j} \equiv \theta_{i,j}(\widehat{\mathcal{P}})$.

Counterfactual method. The key insight of MW is that we can obtain the target counterfactual exclusively from causal effects of monetary policy shocks under the baseline policy $\mathcal{P}$ with only two main assumptions.[13] The two assumptions are (i) the data-generating process is linear and that (ii) the behavior of private agents depends only on monetary policy via

[12] All results use 20,000 posterior draws.

[13] To ease the exposition, we explain the methodology in the context of our application to monetary policy counterfactuals.

the policy interest rate. Intuitively, the latter condition requires that private agents care only about the policy choices and not the specific rule through which these choices arise.[14] Importantly, a broad class of linear DSGE models satisfy these assumptions (e.g., Smets and Wouters, 2007).

The key identification result of MW states that Lucas-critique robust counterfactuals can be constructed as a linear combination of causal effects under the baseline monetary policy rule

$$\widehat{\theta}_{i,j} = \theta_{i,j} + \sum_{k=0}^{\infty} \omega_k \theta^{\mathcal{P}}_{i,k}, \tag{3}$$

where $\{\theta^{\mathcal{P}}_{i,k}\}_{k=0}^{\infty}$ denotes an infinite set of distinct (linearly independent) causal effects of monetary policy shocks, and $\{\omega_k\}_{k=0}^{\infty}$ denotes a corresponding set of scalar weights.

The weights can be chosen by the researcher in order to implement the counterfactual of interest. In our application, we seek to choose weights so that our target variable of interest becomes fully unresponsive to the tariff shock, i.e.,

$$\{\omega_k\}_{k=0}^{\infty} = \text{argmin}_{\{\omega_k\}_{k=0}^{\infty}} \left\| \theta_{z,j} + \sum_{k=0}^{\infty} \omega_k \theta^{\mathcal{P}}_{z,k} \right\|, \tag{4}$$

where index z denotes the target variable (e.g., inflation) and $\|\cdot\|$ denotes some norm (e.g., the Euclidean norm).[15] Given these weights, we can compute the counterfactual response of any variable i directly from (3).

This method is powerful because it allows estimating how tariff shocks propagate under counterfactual monetary policy, while remaining agnostic about many features of the data-generating process. If we were to follow a structural approach instead, we would need to impose many assumptions regarding, e.g., preferences and technologies. The key limitation

[14] This assumption is consistent with high attention to Fed communication to the extent that market participants monitor the Fed exclusively to better forecast future policy rates.

[15] The weights can also be chosen in order to implement a specific policy reaction function (Taylor rule), which constitutes a cross-equation restriction across macroeconomic variables. Thus, a researcher can also set the weights such that a counterfactual policy reaction function holds at any response horizon to obtain the counterfactual under that rule.

of the MW approach is of a practical nature. The exact counterfactual requires an infinite set of distinct monetary policy shock series and associated causal effects, whereas there is only a relatively small set of credibly identified shocks available. Therefore, in practice, we can only compute approximate counterfactuals, as we explain below.

3.3 Implementation

To obtain the effects of tariff shocks and monetary policy counterfactuals, we estimate a tariff VAR and a separate monetary policy VAR. The latter delivers the causal effects of monetary policy shocks used to construct counterfactuals.

Tariff VAR. Based on (1), we estimate a quarterly tariff VAR including nine variables: the trade-weighted average tariff rate, real GDP, CPI inflation, the FFR, real imports, real exports, the terms-of-trade index, real investment, and macroeconomic uncertainty. Except for the inflation, tariff and federal funds rates, all variables are expressed in natural logarithms. The deterministic component includes a constant and a linear time trend, and the lag order is $p = 4$. The estimation sample spans 1990Q1–2024Q4.

The tariff shock is identified based on two assumptions. First, we impose the timing restriction that a tariff shock affects all macroeconomic variables only with a one-period lag, except for the FFR and the tariff rate.[16] In turn, we allow other macroeconomic shocks to affect the tariff rate contemporaneously. This is important because variation in the (trade-weighted) average tariff rate can result from tariff shocks, but may also arise from changes in import composition or differential changes in import prices. By allowing other macroeconomic shocks to impact the tariff rate contemporaneously, we may control for these channels.[17] Importantly, we allow the FFR to respond to tariff shocks contemporaneously, remaining

[16] Formally, let ε_t^j denote the tariff shock and S_j the impact effect, and assume the tariff rate and the FFR are ordered last in y_t. Then, we impose $S_j = (0, \ldots, 0, *, *)'$, where $*$ indicates that we remain agnostic on the respective coefficients.

[17] We emphasize that our VAR is partially identified and we only impose restrictions on the tariff shock but not on other macroeconomic shocks that affect, e.g., supply and demand.

agnostic about the monetary policy response to tariffs, which we eventually perturb in the counterfactuals. Second, we assume partial invertibility of the VAR so that we can recover the tariff shock (Forni, Gambetti, and Sala, 2019). In Section 5.3, we present formal statistical tests and sensitivity exercises that support this assumption.

Monetary policy VAR. We follow MW and estimate a separate VAR model, as introduced in (1), to obtain the causal effects of monetary policy shocks. Although many distinct monetary policy shocks and associated causal effects are desirable to implement the counterfactuals, we use two identified monetary policy shock series, as in MW: the high-frequency-identified shock from Miranda-Agrippino and Ricco (2021, MAR) and the Taylor-rule residuals from Romer and Romer (2004, RR). For maximal consistency, we use the same VAR variables, lag specification, and estimation method as we do for the tariff VAR.

To identify the causal effects of both monetary policy shocks, we follow the internal instruments approach of MW, including both shocks in the VAR vector. The MAR shock is ordered first, and the RR shock is ordered before the FFR, but after all other variables. Identification is achieved via a lower-triangular Cholesky decomposition.[18] The estimation sample is 1969Q1–2014Q4 and is determined by the availability of the monetary policy shocks, which we take directly from MW.[19]

Monetary policy counterfactuals. To construct counterfactuals, we follow the steps explained in Section 3.2, with the following modifications. First, we use our finite set of shocks. Second, we use a weighted Euclidean norm with weights decaying at a quadratic rate when solving the minimization problem in (4). This reflects our preferences as researchers to obtain a counterfactual that is more accurate in the short run.[20] The target variable z is

[18] Ordering the RR shock second-to-last is also done in MW and is often used as "exogeneity insurance" to address residual identification concerns (see, e.g., Ramey, 2016).

[19] We set missing values within the estimation sample to zero. The MAR and RR shocks are available 1980Q1–2014Q4 and 1969Q1–2007Q4, respectively. In Section 5.3, we provide additional results by changing the sample period and using alternative shocks.

[20] Effectively, one can solve the minimization problem by weighted least squares in impulse response space. The baseline sequence of weights is proportional to $(H+1)^2, (H)^2, (H-1)^2, \ldots, (1)^2$, where H is the

either CPI inflation or real GDP. In the context of optimal policy, these counterfactuals map into a loss function that puts a zero weight on output or price stabilization, respectively. As a result, these are hypothetical scenarios that correspond to an extreme policy configuration to illustrate the full range of possible outcomes. Additionally, we consider an intermediate case where the target variable is the FFR, keeping monetary policy unresponsive to the shock.

To implement these counterfactuals, we pick both monetary shocks (the weights) so that the target variable is as unresponsive as possible. This minimization procedure provides us with the weights to compute the counterfactual responses using (3). Finally, for inference, we follow MW and take the baseline response to the tariff shock as given and account for the joint estimation uncertainty of both monetary shocks by solving the minimization in (4) for each posterior draw.

4 Results

This section presents our baseline responses of macroeconomic variables to the identified tariff shock and the corresponding monetary policy counterfactuals.

4.1 The macroeconomic effects of tariff shocks

In Figure 2, we present the baseline responses to a shock that raises the import tariff rate by 1 percentage point on impact. The medians of the posterior distribution are reported as solid blue lines, and the shaded areas indicate 68% and 90% credible sets. As shown in Panel (a), the effects on the tariff rate slowly dissipate over the five-year response horizon.

Core outcomes. Panels (b)–(d) show the responses of CPI inflation, real GDP, and the FFR. Inflation starts increasing with a delay and peaks at 0.78 percentage points after 11 quarters. This inflation effect is somewhat persistent, and the 68% credible set includes zero

maximum response horizon that we report. Using weights with a linear decay yields very similar results.

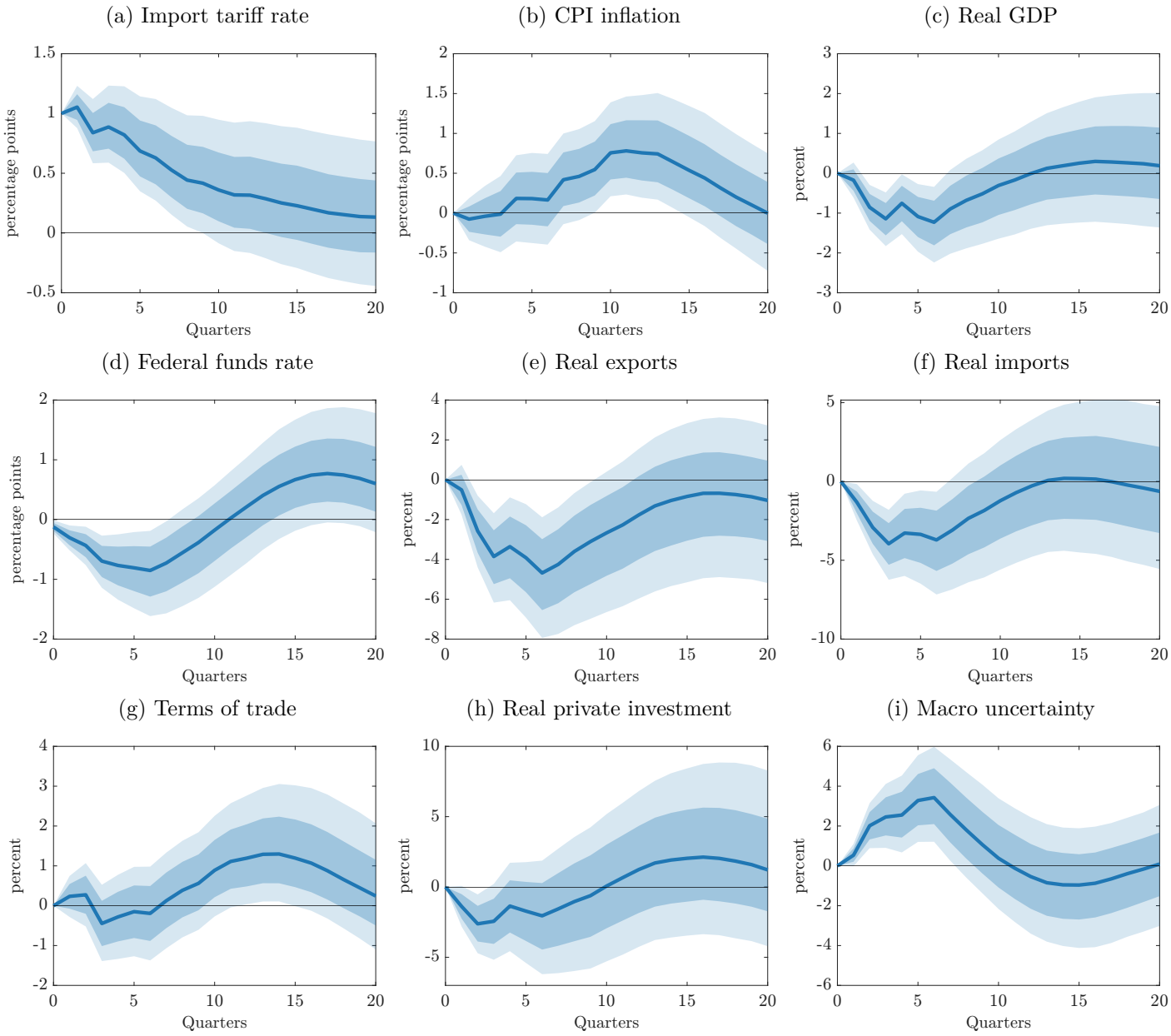
only after 17 quarters. In contrast, real GDP declines more quickly. The trough is reached 6 quarters after the shock, at which output is 1.23% lower. This decline is more transitory and vanishes after the second year, when even the 68% credible set contains zero. Lastly, we find partial monetary accommodation of the tariff shock. The FFR falls for two years. Quantitatively, at the trough, the FFR is 0.85 percentage points lower after 6 quarters and starts to revert thereafter.

Additional outcomes. The responses of the remaining VAR variables are shown in Panels (e)–(i) of Figure 2, providing insight into the broader transmission of tariff shocks. Focusing on trade, we document declining real exports and imports and a more delayed increase in the terms of trade. Macroeconomic uncertainty increases and real investment temporarily declines, consistent with real-option theory.

Discussion. The responses align well with the theoretical literature. First, we confirm that tariff shocks are contractionary, in line with Auclert et al. (2025).[21] Second, tariffs are inflationary and thus act as supply shocks, consistent with Werning et al. (2025). Third, monetary policy partially accommodates contractionary tariff shocks, which cushions the effects on real activity, albeit at the expense of higher prices. Interestingly, such partial accommodation can be the optimal response to a tariff shock (e.g., Bergin and Corsetti, 2025; Bianchi and Coulibaly, 2025; Monacelli, 2025; Werning et al., 2025).

[21] The theory in Antonova et al. (2025) suggests that our tariff shocks mostly capture tariffs imposed on upstream sectors, since they find only upstream sector tariffs to be clearly recessionary.

Figure 2: Responses of macroeconomic variables to an import tariff shock



Notes: This figure shows impulse responses estimated based on a Bayesian VAR, as specified in Section 3. The solid blue line represents the posterior median, and the shaded areas are 68% and 90% credible sets. The responses are identified by imposing the identifying restriction that the shock affects only the import tariff rate and the federal funds rate contemporaneously.

4.2 The monetary policy response to tariff shocks

The result of partial accommodation of tariff shocks by monetary policy raises the question of how these shocks propagate under alternative monetary policy responses. Thus, we present counterfactuals studying scenarios featuring (i) no interest rate response, (ii) full inflation stabilization, and (iii) full output stabilization. We focus our discussion on the counterfactual responses of the core variables, which we present as solid red lines in Figure 3. For comparison, we also show the baseline responses as gray dashed lines.

No interest rate response. A natural benchmark is a scenario where nominal interest rates do not respond, as considered by Auclert et al. (2025). Therefore, we construct a corresponding counterfactual in which the FFR responds as little as possible to the tariff shock. This scenario is shown in the first row of Figure 3. By construction, the FFR is less responsive than in the baseline. Real GDP is broadly unaffected by this alternative monetary response for around 4 quarters. However, the adverse GDP effects are stronger at the trough and more persistent, absent the monetary easing from the baseline. In turn, this results in a peak inflation effect that is 14% (0.11 percentage points) lower, as well as a less persistent inflation response. Thus, we confirm that the inflationary impact of tariffs is partly driven by monetary easing.

Strict inflation stabilization. Since inflation is less persistent absent partial monetary accommodation, we explore how potent monetary policy is in stabilizing prices and what output costs it implies. Thus, we consider a counterfactual in which CPI inflation responds as little as possible and show the results in the second row of Figure 3.[22] This policy requires a short-lived interest rate hike that peaks at 0.83 percentage points only 1 quarter after the shock. Then, the policy rate falls rapidly. Such a sharp and short-lived rate hike is

[22]This counterfactual does not imply zero inflation unconditionally. Instead, it implies that inflation evolves unconditionally as before, except that tariff shocks do not affect inflation. In the hypothetical scenario where no non-tariff shocks affect inflation (or no non-tariff shocks hit the economy), this would imply that inflation is at its long-run value despite tariff shocks hitting the economy.

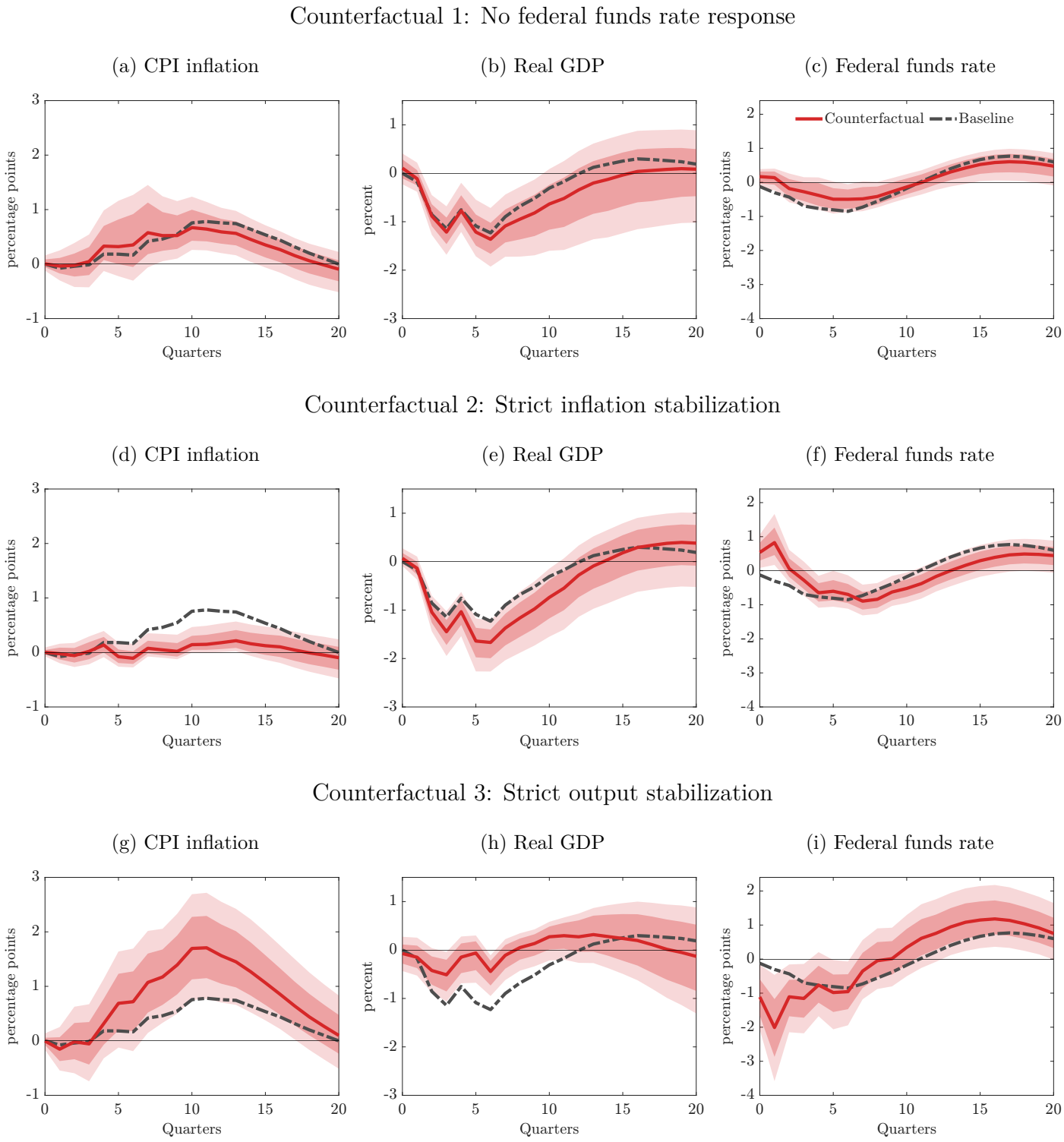
sufficient to tame inflation, with peak inflation reduced from 0.78 to only 0.22 percentage points. However, this policy amplifies the recessionary impact of the shock considerably. The counterfactual GDP trough is 0.44 percentage points lower than the baseline. This represents a 36% increase in the adverse output effects between the baseline and counterfactual troughs. Moreover, it takes 16 quarters for the counterfactual response to catch up with the baseline, suggesting persistent adverse effects.

Strict output stabilization. Lastly, we estimate a counterfactual in which monetary policy aims to fully stabilize real activity. This scenario is displayed in the third row of Figure 3. It is implemented with a peak interest rate cut of about 2 percentage points reached only 1 quarter after the tariff shock. As expected, inflation increases considerably in this counterfactual scenario: the peak inflation effect roughly doubles compared with the baseline (an increase of 0.93 percentage points). This suggests a sizable sacrifice of price stability to minimize the adverse GDP impact.

Additional counterfactual outcomes. We present counterfactual responses of the additional outcome variables in Figure 4. Real investment is an important channel through which interest rates are transmitted. In contrast, macroeconomic uncertainty is less affected, especially in the short run. We also combine the responses of real imports and exports to obtain the effect on the trade balance.[23] Output stabilization reverses the effects on the trade balance, whereas inflation stabilization amplifies the modest trade balance improvement in the baseline. This pattern is consistent with the dominant currency paradigm (Gopinath, Boz, Casas, Díez, Gourinchas, and Plagborg-Møller, 2020), in which muted expenditure-switching effects imply that monetary policy primarily influences the trade balance through domestic demand. Finally, there are no meaningful differences in the terms of trade.

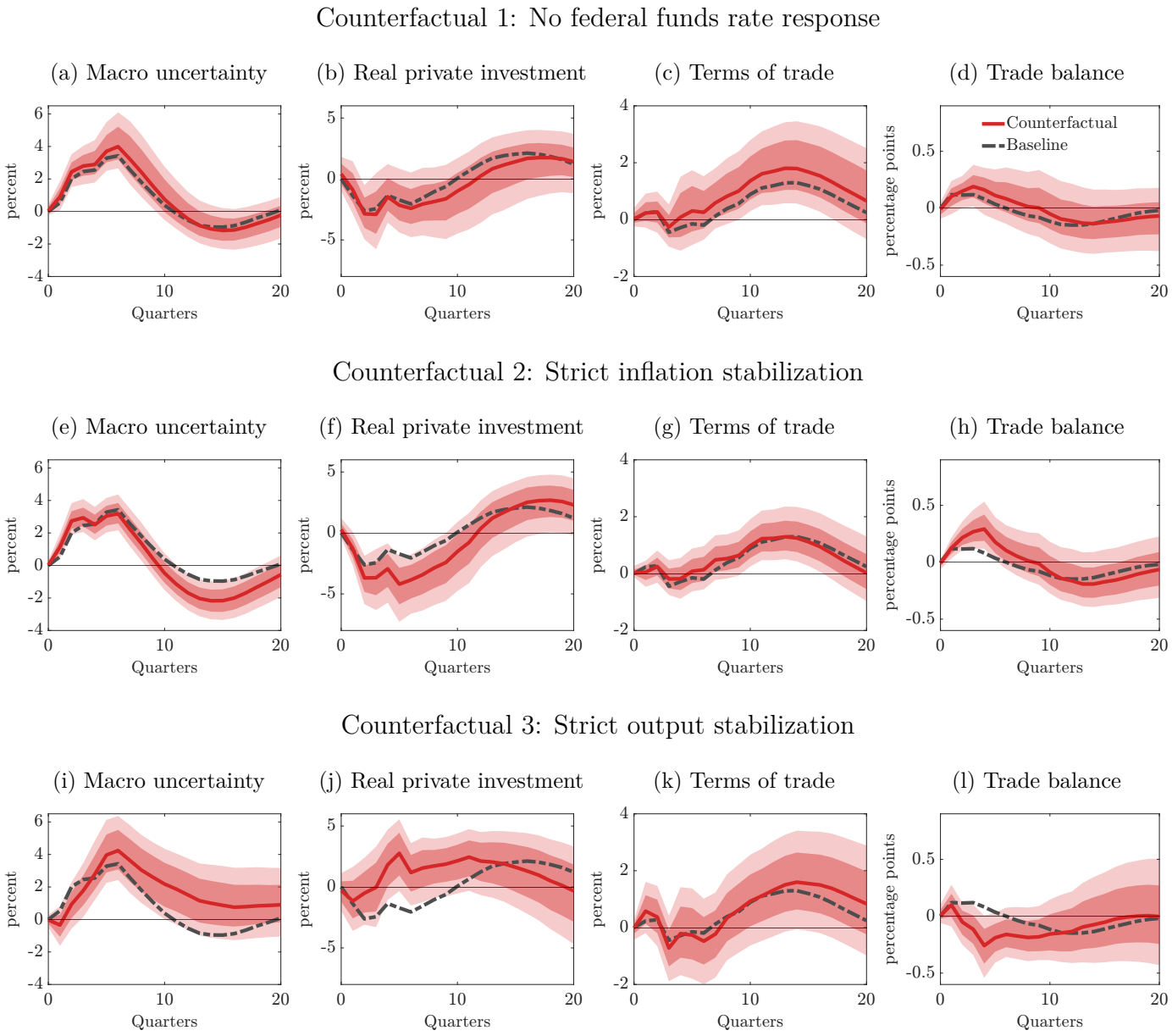
[23] We compute the trade balance response based on our estimated semi-elasticities of real exports and real imports, which we convert into level effects by multiplying by average real exports and average real imports, respectively. We then use the implied level effects to compute real exports minus real imports and divide by average real GDP.

Figure 3: Responses of core variables to tariffs under counterfactual monetary policy



Notes: This figure shows counterfactual impulse responses estimated based on a Bayesian VAR, as specified in Section 3. The counterfactuals are computed using monetary policy shocks following McKay and Wolf (2023). The solid red line represents the posterior median of the counterfactual, and the shaded areas are 68% and 90% credible sets. The dashed gray line corresponds to the baseline median response to a tariff shock, as displayed in Figure 2. For comparison, we keep the vertical axis for each variable across counterfactuals fixed.

Figure 4: Responses of additional variables to tariffs under counterfactual monetary policy



Notes: This figure shows counterfactual impulse responses estimated based on a Bayesian VAR, as specified in Section 3. The counterfactuals are computed using monetary policy shocks following McKay and Wolf (2023). The solid red line represents the posterior median of the counterfactual, and the shaded areas are 68% and 90% credible sets. The dashed gray line corresponds to the baseline median response to a tariff shock, as displayed in Figure 2. For comparison, we keep the vertical axis for each variable across counterfactuals fixed.

5 Additional results and sensitivity analysis

5.1 The role of the sample period

Differences from the related literature. Two closely related papers are Schmitt-Grohé and Uribe (2025, henceforth SGU), who study permanent and transitory tariff shocks, and den Besten et al. (2026, henceforth dBBKS), who present a long-run narrative analysis. Their results agree that non-permanent tariff increases are deflationary post-World War II, but only dBBKS find tariffs to be contractionary. Although our paper and the two aforementioned works differ in a range of aspects, including modeling choices and identification approaches, we find that the sample period plays an important role in explaining the differences from our work.

SGU use quarterly data covering 1959Q2–2024Q4, whereas dBBKS use annual data covering 1946–2024 for their post-World War II analysis. In contrast, we use a 1990Q1–2024Q4 sample, which we view as desirable because it is more representative of the current economic environment and less susceptible to structural breaks. For example, increasing supply chain complexity and the growing importance of intermediate inputs may alter the effects of tariffs on domestic prices (Bergin and Corsetti, 2023; Moro and Landi, 2026).

Reconciliation with the related literature. In Online Appendix B, we present results using longer samples. Tariffs remain contractionary and inflationary if we expand the sample to start in 1980. When extending the sample back to 1967, the earliest year permitted by data availability, tariffs instead appear deflationary and not contractionary. These deflationary effects seem relatively robust in longer samples, consistent with SGU and dBBKS, and with intermediate inputs being less important in the first half of this extended sample (Bergin and Corsetti, 2023). However, the absence of an output contraction in the extended sample is exclusively driven by two extraordinary tariff spikes during the Nixon and Ford Administrations, both of which are classified as endogenous events in the narrative account

of dBBKS. Thus, one cannot use these events to identify the causal effects of tariffs. If we exclude these events, we recover a significant real GDP contraction.

5.2 Additional outcome variables

In Online Appendix C, we present the responses of several additional variables.

Alternative core variables. We consider a range of alternative measures for real activity (unemployment rate, output gap, and industrial production), prices (PPI inflation, core CPI inflation, and one-year inflation expectations), and interest rates (one-year Treasury yield and the shadow FFR). Across all specifications, we find that tariffs are contractionary, inflationary, and accompanied by lower nominal rates.

Fiscal implications. We find that tariff shocks generate customs revenue and induce a corresponding transitory increase in government spending, consistent with the narratives of tariff proponents. However, total government receipts tend to decline due to the contractionary impact. Ultimately, we find that tariffs have a negative effect on the government budget, and federal debt increases.

Trade balance. Tariffs are often justified with protectionist arguments, claiming that they shield domestic producers from foreign competition and thereby improve the trade balance. If anything, we find that the trade balance improves only transitorily, indicating little evidence for the protectionist view.

Consumption. We also study the effects of tariffs on private consumption. Private consumption declines strongly and broadly mirrors the response of real GDP, whereas consumer goods imports experience a more temporary decline.

5.3 Sensitivity analysis

In Online Appendix D, we investigate the sensitivity of our results to various modeling choices, finding that our key insights are remarkably robust.

Anticipation of tariff changes. Tariffs may be announced ahead of implementation. Such tariff news shocks could lead to a failure of invertibility, a key identifying assumption. If this were a substantial concern, we should expect that lagged financial market data and professional forecasts would predict our shock. However, neither net-export forecasts from the SPF nor principal components from a large macro-financial dataset have significant predictive power for the tariff shock.[24] The latter exercise is precisely the procedure from Forni and Gambetti (2014) to formally test for invertibility failures. Lastly, our results are also robust to including several of the principal components, which comprehensively controls for anticipation effects.

Relaxing identifying assumptions. We evaluate the baseline zero restrictions by considering three alternatives. First, we employ a block-recursive identification similar to Christiano, Eichenbaum, and Evans (2005): we relax the zero restrictions on all "fast-moving" variables, namely the terms of trade, macroeconomic uncertainty, and inflation. Second, we relax all zero restrictions using the penalty function approach of Uhlig (2005): we identify the tariff shock as the one that increases the tariff rate the most over the first four response horizons. Third, we include a series of narratively identified tariff events as an exogenous variable in the VAR. Regardless of which approach we take, we obtain estimates similar to the baseline.

Alternative tariff measures. Our baseline tariff measure is trade-weighted, which is desirable because goods-level tariffs are weighted by aggregate importance. However, trade

[24] Net-export forecasts are an indirect proxy for tariff expectations, which we use because we do not observe the latter.

weights can respond to non-tariff shocks, creating additional identification concerns. When using the unweighted tariff rate, we obtain broadly similar but less significant results. This reduction in significance is expected because tariffs are not weighted by their aggregate importance. Finally, we use the trade restrictiveness index from Schmitt-Grohé and Uribe (2025), which yields larger magnitudes, possibly because it also captures distortions due to cross-sectional tariff variation. In both cases, we still find that tariffs raise prices but depress output.

Modeling choices. We focus on a range of additional modeling aspects. First, we relax VAR assumptions by adjusting the lag order or estimating local projections, as recommended by Montiel Olea, Plagborg-Møller, Qian, and Wolf (2025). Second, we account for the Covid-19 pandemic by dummying out the pandemic period (Lenza and Primiceri, 2022). Third, we add higher-order deterministic time trends to account for slow-moving trends in international trade. Fourth, we include a wide range of policy uncertainty measures in the VAR. None of these extensions change our conclusions.

The role of monetary policy shocks. The counterfactuals hinge on the monetary policy shocks used. Thus, we first study their raw impulse responses. Both shocks yield responses that align well with theoretical predictions. Second, to understand the gains from including two shocks, we re-estimate all counterfactuals using only a single shock. Although the results are qualitatively similar, we find that the precision with which we can implement the counterfactuals is considerably higher with two shocks. Third, we vary the sample period and use alternative shocks identified using a heteroskedasticity-based approach (Jarociński, 2024) and an augmented Romer and Romer (2004) regression that accounts for time variation in systematic monetary policy (Hack et al., 2024). Throughout, we obtain results that are similar to the baseline.

6 Conclusion

This paper provides two main empirical insights. First, tariffs act as supply shocks, which monetary policy partially accommodates by lowering interest rates. Second, the macroeconomic effects of tariffs are strongly shaped by systematic monetary policy.

Finally, we note that the effects of tariffs may depend on many aspects, including the degree of persistence, the credibility of tariff announcements, and the composition of tariffed goods. These aspects possibly set tariffs in our sample apart from tariffs imposed by the Trump Administration in 2025. Thus, we caution against applying our results directly to calculate the impact of 2025 tariffs. Instead, future research may combine our estimates with a structural model to account for the above-mentioned aspects of tariffs, e.g., using a multi-sector model. Going beyond positive analysis, such a model could also be used to assess the optimal monetary policy response to tariffs. Beyond monetary policy, future work may also consider fiscal policy counterfactuals that vary how tariff revenues are used, which may complement existing theoretical analysis (Alessandria et al., 2025).

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